

AARUSH PANDA

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EDUCATION

University of California, Berkeley | Expected May 2027

Berkeley, CA

B.S. Mechanical Engineering (Aerospace Concentration), Certificate in Entrepreneurship & Technology

Major GPA: 3.56

Involvements: Undergraduate Research, VTOL @ Berkeley; Captain, Division I Club Frisbee Team

PROFESSIONAL EXPERIENCE

Naval Nuclear Laboratory - Bettis Atomic Power Laboratory (DOE L Clearance)

Pittsburgh, PA

Mechanical Engineering Intern

Jun 2025 – Aug 2025

- Designed and tolerance-stacked a payload-leveling mechanism for spent-fuel exams, achieving $\pm 0.1^\circ$ levelling accuracy
- Evaluated structural integrity via hand calculations and MathCAD analyses, confirming safety factors against design loads
- Engineered ASME Y14.5-compliant Creo assemblies, improving assembly repeatability across maintenance cycles
- Implemented DFM changes to reduce feature complexity and tolerance variation, cutting prototype lead time by 25%

VTOL (Vertical Takeoff and Landing) @ Berkeley

Berkeley, CA

President and Lead Mechanical Engineer

Aug 2023 – Dec 2025

- Led end-to-end mechanical design of Berkeley's 1st human-carrying eVTOL aircraft, delivering flight-critical hardware
- Designed, integrated, and tested carbon-fiber landing gear and tiltrotor mechanisms in SolidWorks and Fusion 360
- Calculated FAA-compliant load cases and verified structural integrity with safety factors under worst-case loads
- Optimized carbon-fiber structures through FEA and prototyping, reducing mass by 15% and improving thrust-to-weight

Berkeley Risk and Security Lab

Berkeley, CA

Undergraduate Researcher

Sep 2024 – Present

- Performed stress and fracture analyses on pressure vessels across failure modes, guiding non-proliferation material choices
- Mapped structural load paths across reactor assemblies to identify load-bearing components constraining physical access
- Analyzed failure modes of multi-sensor counter-UAS systems supporting reliable operation under contested conditions

ENGIE North America

Livermore, CA

Energy and Sustainability Intern

Jun 2023 – Aug 2023

- Optimized lighting, solar, and HVAC energy models, saving 718,000+ kWh and \$178K annually (27% reduction)
- Built LEED BD+C plan aligning IEQ and renewable credits, securing ENGIE and board approval
- Communicated technical analyses through articles, presentations, and videos, educating 1,500+ students on energy topics

PROJECTS

Custom Tilt-Rotor UAV ([Link](#))

- Led end-to-end mechanical and aerodynamic design, build, and test of a custom tiltrotor UAV
- Performed ANSYS CFD across hover and cruise conditions, reducing wake interference and drag
- Designed and fabricated carbon-fiber primary structures and tilt mechanisms, validating strength under operational loads
- Tuned ArduPilot flight controls and validated performance via flight testing, achieving stable hover-to-cruise transitions

Robotic Shoe Shelver System ([Link](#))

- Designed automated load-cell triggered actuator system enabling hands-free shoe removal
- Integrated load cell, actuator driver, and sensor feedback to prevent overload and repeated actuator-induced failures
- Built event-driven MicroPython and ESP-NOW logic enforcing safe sequencing between electromechanical subsystems

HushLight (In Progress)

- Designed and built compact electromechanical system integrating sensing, actuation, and visual feedback
- Engineered enclosure geometry and mounting interfaces to support constrained packaging, durability, and assembly
- Assembled and bench-tested functional prototype, validating threshold-based activation and repeatable system robustness

Hardware Analysis for NASA and DoD Systems (In Progress)

- Developed hardware-centric classification framework based on component function, lifecycle phase, and failure criticality
- Analyzed how hardware design decisions affect failure modes, maintainability, and operational readiness
- Applied classification insights to identify key drivers of readiness, maintenance delays, and integration bottlenecks

SKILLS & INTERESTS

Hardware: Structural Analysis, FEA, GD&T, DFM, DFMEA, Additive Manufacturing (FDM/SLA), Machining, Composites

Software: SolidWorks, Creo, Fusion 360, ANSYS (Structural/CFD), MATLAB, Python, MathCAD, Windchill PLM

Interests: Ultimate Frisbee, Soccer, Defense Strategy, Weightlifting, Entrepreneurship